

The geothermal system of the San Pedro Dome constrained by magnetotelluric and lithological information

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ABSTRACT

The San Pedro Dome geothermal field is located on the northeastern limit of the Compostela graben, at the western of the Tepic-Zacoalco rift, southwest of the Nayarit state in Mexico. Despite detailed geophysical surveys around the San Pedro Dome conducted by private companies to locate the permeable hydrothermal reservoir, very little is known about the heat source and structures that conform to the geothermal system. To investigate the role of the identified geoelectric structures in the region and the general distribution of the geothermal reservoir, we inverted the full impedance tensor of a set of 11 magnetotelluric soundings in the range of 10^{-2} to 10^3 Hz using a 3-D algorithm. The regional resistivity model was constrained with available information of geothermal wells, surface geology, and existing regional gravity and magnetic data. The results include a 3-D resistivity image of the subsurface structure, useful to

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distinguishing moistened from dryer zones within the upper and lower crustal granitic basement rocks of the studied area. The interpreted 3-D model shows the existence of a horizontal conductive layer of variable thickness located at an elevation of 0 m and less than 1 km of thickness, which is identified as the confining cap layer of the San Pedro Dome geothermal system, and which is separated from other cap layers related to other promising geothermal centers, such as the Amado Nervo volcano, southward from the interest area within the Compostela graben.

Keywords: magnetotellurics, lithosphere, Tehuantepec, transform, fault, Chicxulub

RESUMEN

El campo geotérmico del Domo de San Pedro (DSP) se localiza en el límite nororiental del graben de Compostela, al oeste del rift de Tepic-Zacoalco, en el suroeste del estado de Nayarit, México. A pesar de los estudios geofísicos detallados realizados por empresas privadas en dicha zona de estudio, orientados a la localización de reservorio hidrotermal permeable, aún se desconoce acerca de la fuente de calor y de las estructuras que conforman el sistema geotérmico del DSP. Con el propósito de investigar el papel de las estructuras geoelectricas de la región y la distribución general de dicho reservorio geotérmico, se invirtió el tensor de impedancia completo de un conjunto de 11 sondeos magnetotéluricos, en el rango de 10^{-2} a 10^3 Hz, mediante un algoritmo tridimensional. El modelo regional de resistividad eléctrica se ajustó con la información disponible de pozos geotérmicos, la geología superficial y datos gravimétricos y magnéticos regionales existentes. Los resultados proporcionan una imagen tridimensional de la resistividad eléctrica del subsuelo, que permite distinguir distintas zonas (húmedas y secas) dentro de las rocas graníticas que conforman el basamento cortical superior e inferior del área estudiada. El modelo 3D interpretado revela la presencia de una capa conductora horizontal de espesor variable, situada aproximadamente a una elevación de 0 m sobre el nivel del mar y con un espesor menor a 1 km. Esta capa se interpreta como el nivel de confinamiento del sistema geotérmico del DSP, el cual se encuentra separado de otras capas conductoras asociadas a potenciales centros geotérmicos, como es el caso del Volcán Amado Nervo, localizado al sur del área de estudio dentro del graben de Compostela.

Palabras clave: magnetotelúrica, litosfera, Tehuantepec, transformada, falla, Chicxulub

1. INTRODUCTION

The Compostela graben is the westernmost structure that conforms to the Tepic-Zacoalco rift, which is the limit between the Jalisco Block and Sierra Madre Oriental (Fig. 1). The San Pedro Dome (SPD) is located on the northeastern boundary of the Compostela graben, which in turn is a consequence of the extensional tectonic regime of the Tepic-Zacoalco rift in this particular area.

The geophysical prospection in the SPD started 10 years ago, initially focusing on identifying suitable locations for productive geothermal wells. Following two costly failures primarily based on surface geochemical and physical evidence, the strategy changed, and 30 audio-magnetotelluric soundings were carried out in a specific spot area, constrained by detailed structural and

geologic studies. The combined results pointed toward the current production zone of the SPD (red spots in Fig. 2), where the first well drilled was highly productive. Several more productive wells have been drilled in this area, contributing to the known geothermal energy potential of the country. Nearly 120 additional magnetotelluric (MT) soundings were acquired, mainly in the N-NW of the SPD to better define the structure of the geothermal field. However, these results and data are not public; the current installed capacity of the SPD geothermal field in 2020 was of 35.5 MW (Orozco et al., 2020).

Analysis of drilled well stratigraphy identifies four main geological formations, spanning from the late Pliocene and Quaternary to the Cretaceous-Paleogene basement rocks (Orozco et al., 2019), each with different degrees of alteration. Nevertheless, the depth of the wells (<3 km) is insufficient to shed light

on the extension of the reservoir at depth, the heat source of the field, and the pathways of pressurized steam toward the surface. Until now, it is unknown if the higher-enthalpy, higher-porosity distribution in the surroundings of the SPD depends on a regional view of the distribution of structures. The relationship between the SPD geothermal field and the Compostela graben and whether

there are other suitable areas for geothermal development within it are also unknown.

To give light on the regional structure of the SPD geothermal field and the volcanic environment where it is immersed, the present study provides a working calibrated model of the structural configuration and general physical characteristics of the SPD.

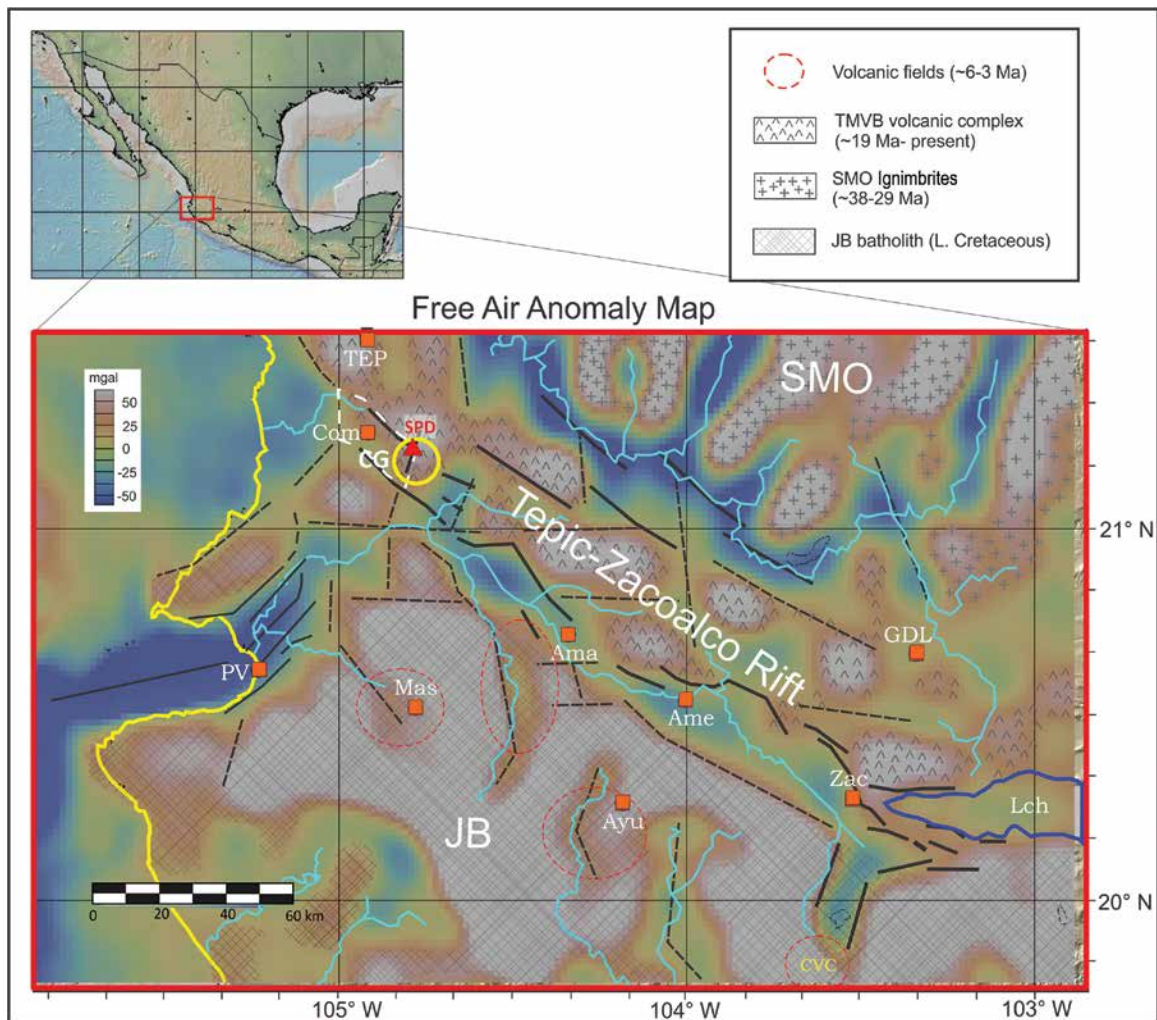


Figure 1. Free air anomaly map (Ryan et al., 2009; <https://www.geomapapp.org>) covering the northern structural limits of the Jalisco Block (JB) granitic complex and the active Tepic-Zacoalco Rift which contains the volcanism of the Trans-Mexican Volcanic Belt (TMVB) and separates the JB from the Sierra Madre Occidental (SMO) ignimbrite province. The yellow circle represents the analyzed area in this work. The red triangle is the San Pedro Dome (SPD), the dotted white line corresponds to the Compostela graben (CG), and the red dotted ellipses mark the volcanic fields (6–3 Ma) of Mascota (Mas), Amatlán (Ama), Ayutla (Ayu) and Colima (CVC), mainly located within the JB. Other cities and towns (orange squares) are Ameca (Ame), Compostela (Com), Guadalajara (GDL), Puerto Vallarta (PV), Tepic (TEP), and Zacoalco de Torres (Zac). Lch is the Chapala Lake.

Figura 1. Mapa de anomalías de aire libre (Ryan et al., 2009; <https://www.geomapapp.org>) que abarca los límites estructurales septentrionales del complejo granítico del Bloque de Jalisco (JB) y del Rift de Tepic-Zacoalco, el cual contiene el vulcanismo del Eje Volcánico Transversal (TMVB) y separa el JB de la provincia de ignimbritas de la Sierra Madre Occidental (SMO). El círculo amarillo representa el área analizada en este trabajo. El triángulo rojo corresponde al Domo de San Pedro (SPD), la línea blanca punteada al graben de Compostela (CG), y las elipses rojas punteadas marcan los campos volcánicos (6–3 Ma) de Mascota (Mas), Amatlán (Ama), Ayutla (Ayu) y Colima (CVC), ubicados principalmente dentro del JB. Otras ciudades y pueblos (cuadrados naranjas) son Puerto Vallarta (PV), Compostela (Com), Tepic (TEP) y Guadalajara (GDL). Lch es el Lago de Chapala.

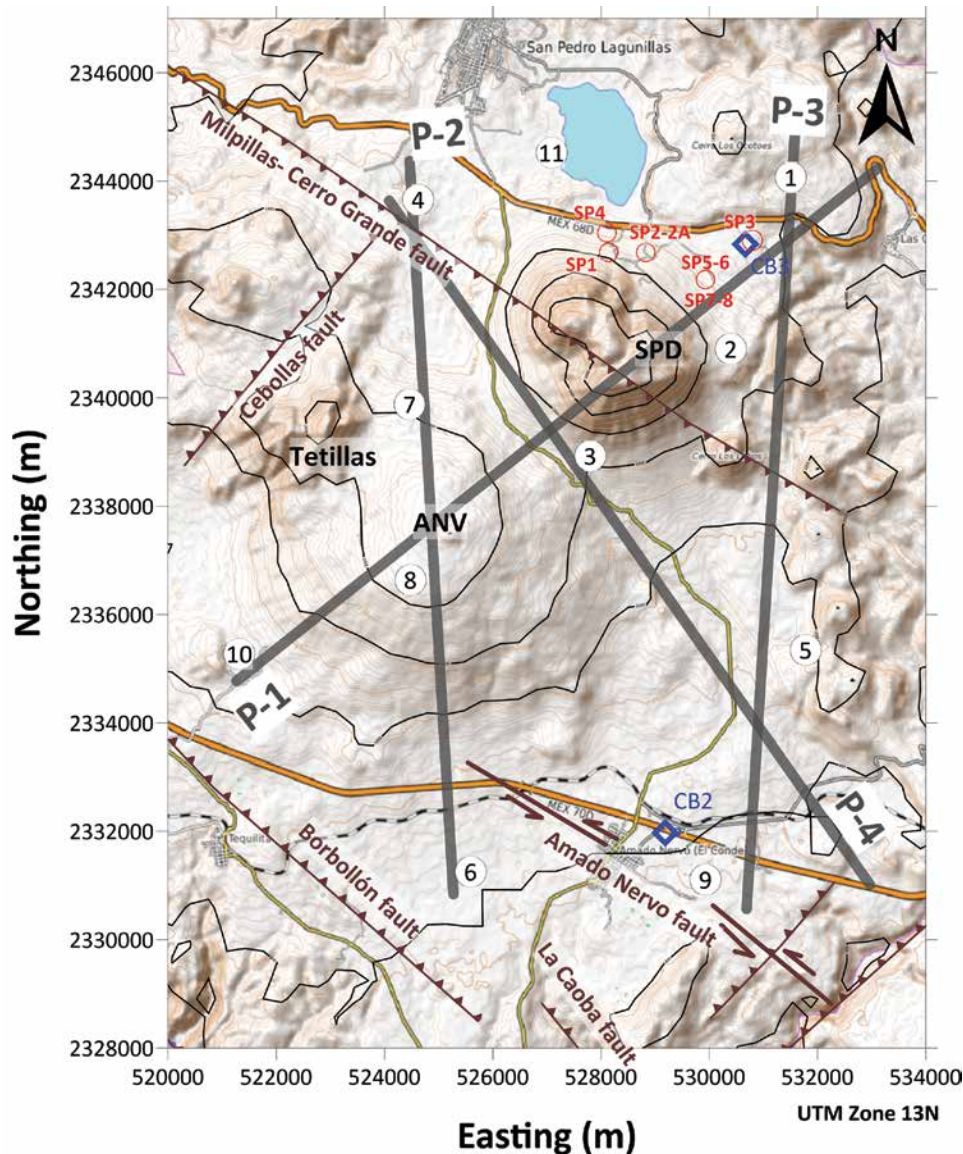


Figure 2. Location of the 11 magnetotelluric stations considered for the 3-D inversion modeling. Sites are shown as numbered circles, geothermal wells as red circles (Orozco et al., 2019, 2020); blue diamonds are exploratory dry wells of the Comisión Federal de Electricidad (Ferrari et al., 2003). Mapped faults by the Servicio Geológico Mexicano (SGM, 2006a) are shown in brown as continuous barbed lines. SPD—San Pedro Dome; ANV—Amado Nervo volcano. Gray thick lines show profile sections extracted from the 3-D resistivity model.

Figura 2. Ubicación de las 11 estaciones magnetotéluricas consideradas para el modelo de inversión 3D. Los sitios se muestran como círculos numerados; los pozos geotérmicos como círculos rojos (Orozco et al., 2019, 2020) y los pozos exploratorios de la Comisión Federal de Electricidad (Ferrari et al., 2003) como rombos azules. Las fallas mapeadas por el Servicio Geológico Mexicano (SGM, 2006a) se muestran como líneas continuas en color marrón. SPD—Domo San Pedro; ANV—Volcán Amado Nervo. Las líneas grises muestran las secciones extraídas del modelo 3D de resistividad eléctrica.

This is useful as a reference for further geothermal development in the area and to fulfill cities' energy needs. This geophysical work must be considered in the context of future monitoring of this active geothermal field to observe and understand its evolution in the measured proxies. The generated knowledge of the full geothermal system in a volcanic zone would be an opportunity to understand the behavior of active and dormant volcanos and their relationship with the upward migration of mineralized hot fluids and high-pressurized steam.

2. SPD TECTONIC SURROUNDINGS

The Tepic-Zacoalco rift forms a nearly 400-km-long NW-SE-trending tectonic boundary between the Sierra Madre Occidental to the north and the Jalisco Block to the south (Fig. 1). The NW end of the Tepic-Zacoalco rift met at a right angle with

the Bahía de Banderas Graben, while the SE end joins the Graben de Colima rift at the Guadalajara Triple Point. The time-elapsed extensional process triggered at ca. 20 Ma (Rosas-Elguera et al., 2003) has led to a segmented Tepic-Zacoalco rift consisting of separated graben and semi-graben structures containing or neighboring with isolated volcanic fields as recently as the end of the last ice age. The Compostela graben, bounded by normal fault is a complex extensional NW-SE basin, located at the westernmost end of the Tepic-Zacoalco rift, and with the same orientation. It has been defined as a Neogene complex extensional basin trending WNW-ESE, developed in the northernmost part of the Tepic-Zacoalco rift in western Mexico (Ferrari et al., 2003). In western Mexico, the production of the volcanism is related to the subduction of the Rivera and Cocos plates beneath the North American plate along the Middle American Trench, but also to the detachment of the Baja Peninsula that started with the Neogene.

The Compostela graben is bounded to the north by the Milpillas–Cerro Grande fault and to the south by the Borbollón fault. The SPD is located on top of the northern fault where the stratigraphy is known from the drilled wells, encountering: dacitic and rhyolitic flows (0.6–0.1 Ma) with thickness above 700 m; 150 m of andesites and dacites of the Pliocene; andesites, basaltic andesites, and basalts of the Miocene; and granodiorite and diorite from the late Paleocene and upper Cretaceous, associated to the basement of the Jalisco Block, which is bounding to the south of the Compostela graben.

The role of faulting in relation to the geothermal reservoir is not fully understood in this extensional volcanic context. The Milpillas–Cerro Grande fault acts as a retention barrier of the SPD geothermal field, but there must be highly permeable vertical paths driving fluids at high pressures. Mapping the fractured granitic basal rocks is therefore of key importance in assessing the geothermal potential resources of the volcanic region within the Compostela graben.

3. THE MT DATA AND 3-D MODELING

The MT method has been successfully employed for geothermal exploration as a main tool in the prospecting phase previous to the full development of the resource (Spichak and Manzella, 2009; Cumming, 2009; Piña-Varas et al., 2014; Muñoz, 2014; Arzate et al., 2018; Corbo-Camargo et al., 2020; Guevara et al., 2020; Pereyra Quiroga et al., 2023). This work presents the results obtained with 11 broadband MT soundings acquired mainly within the Compostela graben (Fig. 2). The data were collected using Phoenix MTU-5A equipment and proprietary software to process the data.

Phase Tensor

Before conducting formal MT data inversion, certain geoelectric features of the SPD are observed in the MT responses. One transfer function that describes the relationship between the electric and magnetic fields is the phase tensor (PT; Caldwell et al., 2004; Booker, 2014), which is independent of galvanic distortions caused by near-surface heterogeneities. The PT is represented as an ellipse as a function of the period for each station in a plane or along a profile (Fig. 3), and the magnitude of the principal axes is a function of the maximum and the minimum value of the PT, while the skew (angle β) defines the degree of symmetry. Therefore, the ellipticity indicates the dimensionality of the structure in the subsurface, whether it is 1-D and/or 2-D/3-D; if β is ~ 0 , the medium is 1-D or 2-D. Also, a PT value $>45^\circ$ suggests a transition to a more conductive medium (hot colors in Fig. 3). Values $<45^\circ$ (cold colors in Fig. 3) imply that the resistivity increases with depth.

In Figure 3, in the range of frequencies above 10 Hz (left panel), the PT exhibits a circular shape (Fig. 3B) and shows low β -angles (Fig. 3D), suggesting a 1-D structure. However, at some higher frequencies (above 100 Hz) in MT soundings 1–4,

the β -angles are elevated, likely due to distortive effects from inductive phenomena caused by topography. The phases $>45^\circ$ indicate a transition to a more conductive medium.

Between 10 Hz and one second, there is a transition from conductor to resistor featuring phases below 45° . This transition is interpreted as for those frequencies, the induction volume has encountered the presence of the granite basement of the Jalisco Block complex. The ellipses show high dispersion in terms of elongation and orientation but maintain low β -angles, suggesting the presence of faults and fractures. Below one second, the β -angle increases in all MT soundings, indicating a 3-D medium. At high periods greater than 10 s, the phases suggest the presence of a more conductive medium, possibly indicating the brittle-ductile transition zone.

Inverse Modeling

One-dimensional modeling of the apparent resistivity and phase curves as a function of frequency was performed to obtain the average conductivity structure of the subsurface, by inverting the Berdichevsky invariant of Z. Figure 4 shows a comparison of Occam 1-D models (Constable et al., 1987) for MT-1 and MT-2, which are located close to the wells SP3 and SP7–8, respectively. Both wells reach the upper part of the granite, characterized by its high electrical resistivity. Within the upper 1,000 masl, a conductive layer correlates with altered andesites and dacites that forms an impermeable seal for pressurized fluids. Toward MT-2, the thickness of the seal increases and becomes more conductive (~ 5 Ohm-m). Overlaying the conductive seal lies a layer of unaltered resistive rhyolitic lava flows from the later volcanic activity of the SPD (Ferrari et al., 2003; Orozco et al., 2019).

The less resistive the granite is at depth, the more fractures and fluids it may have. High conductivities in a resistive rock matrix are largely related to the highly mineralized water and to the permeability distribution of the rock which hosts the hydrothermal reservoir. The natural electromagnetic fields on which the MT method is based are used to trace pathways along contact surfaces or faulted terrain where hot mineralized fluids remain trapped during dormant periods beneath the conductive sealing clay cap layer. This layer is primarily characterized by smectite minerals at temperatures ranging from 70 to 150 °C, and by precipitated magnetite and ferromagnesian minerals at higher temperatures ranging from 150 to 300 °C (e.g., Jording et al., 2000; Cumming and Mackie, 2010; Corbo-Camargo et al., 2020; Hering et al., 2021; Arzate et al., 2023).

We inverted the full impedance tensor of the MT soundings using the 3-D inversion algorithm ModEM (Egbert and Kelbert, 2012; Kelbert et al., 2014). The model mesh comprised 51 cells of 800 m of lateral extension covering the survey area. Vertically, the mesh consisted of 42 cells with an increasing thickness by a factor of 1.2 from an initial value of 50 m. Padding horizontal cells of 15×15 increment of a factor of 1.3 were included to avoid the boundary effects (Fig. 5A). The mesh included the topography to account for inductive distortion (Jiracek, 1990;

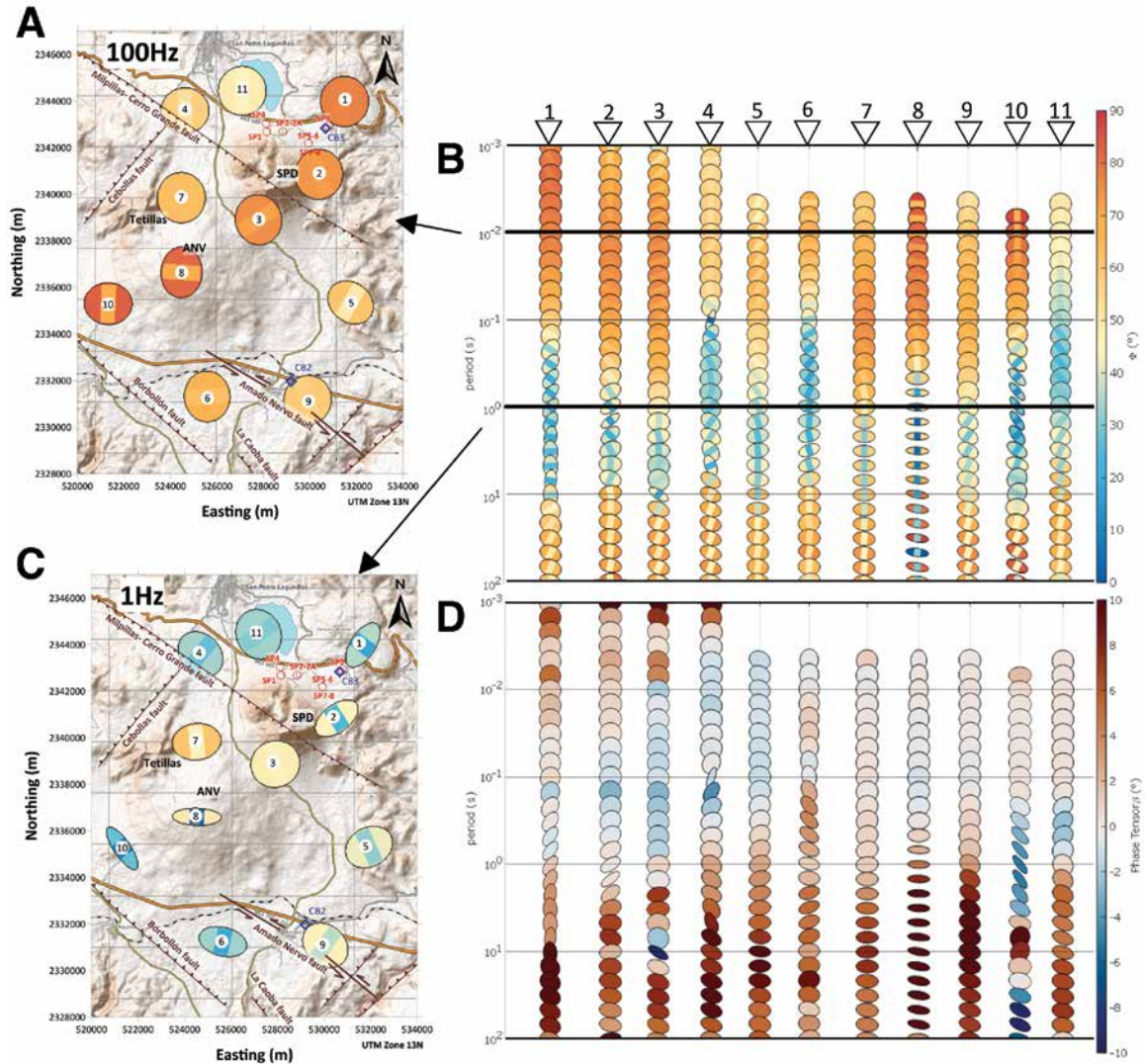


Figure 3. Phase tensor (PT) maps for all sites and periods (B) with floor plans at 100 Hz (A) and 1 Hz (C). PT with angle β is presented in (D). SPD—San Pedro Dome; ANV—Amado Nervo volcano.

Figura 3. Mapas del tensor de fase (PT) para todos los sitios y períodos (B), planos de planta a 100 Hz (A) y 1 Hz (C). El PT con su ángulo β se presenta en (D). SPD—Domo de San Pedro; ANV—Volcán Amado Nervo.

Nam et al., 2008). The inversion process was performed multiple times using different combinations of parameters. The preferred inversion model was obtained using a starting homogeneous half space of 100 Ohm-m for land and 0.3 Ohm-m for the ocean layer, which is kept fixed in the inversion mesh. Using a covariance factor of 0.2, the inversion process converged after 143 iterations with a normalized root mean square of 1.62. Data error and error floors of 5% of $|Z_{xy} \cdot Z_{yx}|^{1/2}$ were assigned for all complex components. Figure 5C shows the model grid and the resulting 3-D resistivity model viewed from the southwest. Fitting curves for all components are presented as Supplemental Material Fig. S1¹.

3-D Inverse Modeling Test

To analyze the sensitivity of the preferred inversion models, we compared them with information from the wells in the SPD, achieving a very satisfactory concordance (Fig. 4), particularly in the conductive layer indicated by the clay cap and the increase in electrical resistivity due to the upper part of the Jalisco Block complex granite. Furthermore, the PT response

¹Supplemental Material. Figures S1–S4 and Supplemental video (Movie.wmv). Please visit <https://doi.org/10.1130/SPE.S.32642913> to access the supplemental material, and contact editing@geosociety.org with any questions.

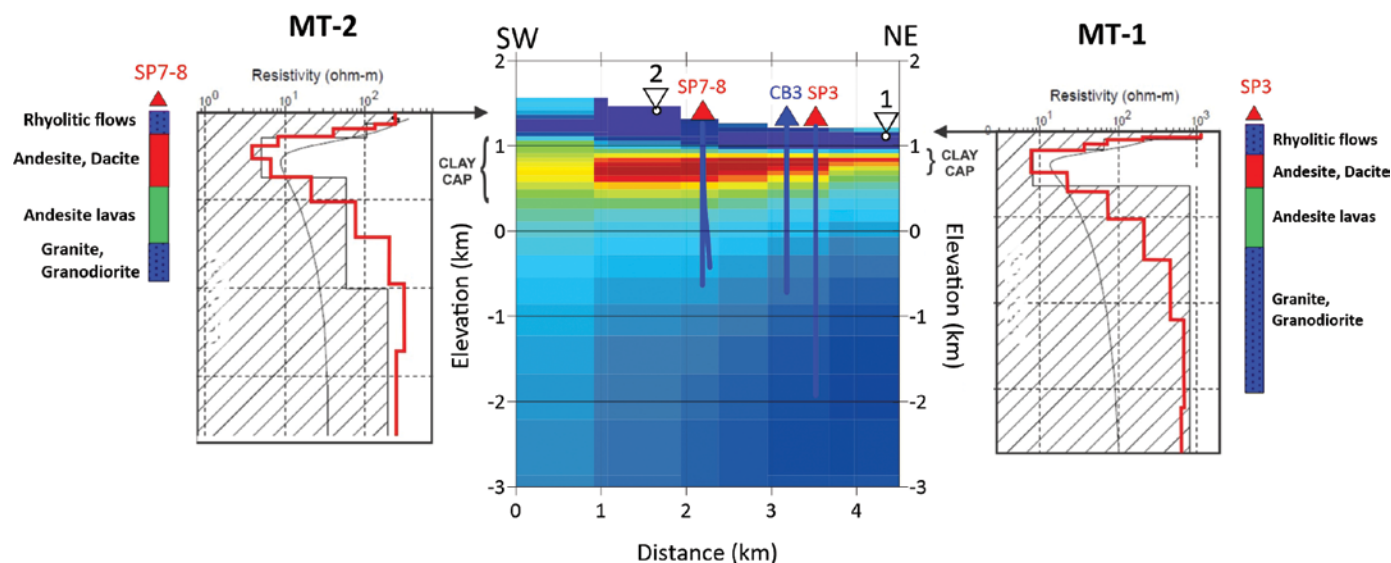


Figure 4. Comparison of Occam 1-D resistivity models as a function of soundings MT-1 and MT-2 with the nearest wells SP3 and SP7-8, respectively. Both wells reach the upper part of the granite, distinguished by its high electrical resistivity. The central part of the figure shows a fragment of the 3-D inversion results along profile P-1 (Fig. 8A).

Figura 4. Comparación entre los modelos 1D de resistividad de Occam de los sondeos MT-1 y MT-2 con los pozos más cercanos SP3 y SP7-8, respectivamente. Ambos pozos alcanzan la parte superior del granito, caracterizada por su alta resistividad eléctrica. En la parte central de la figura se presenta un fragmento del resultado de la inversión 3D a lo largo del perfil P-1 (Fig. 8A).

showed these transitions, indicating the presence of fractures and faults in the granites. Based on this, several tests were conducted to determine the response to potential changes in electrical conductivity in the Jalisco Block, which is indicative of weakness zones (fractures, faults) through which hot fluids may rise. To test the sensitivity, we performed additional 3-D inversions, with changes in the resistivity of the model starting at 0 masl (Fig. 6), which we consider as an equilibrium line of the Jalisco Block granitic complex.

First, a homogeneous half-space of 25 Ohm-m (Fig. 6C) was considered, which is significantly lower than the resistivities obtained for the Jalisco Block. The forward responses were calculated to compare with the observed MT curves (Fig. S2). The misfit from 10 Hz indicated that the model requires higher resistivities from the Jalisco Block to replicate those from observed data. Subsequently, a 3-D inversion was executed using the same parameters as the original inversion. The result obtained after 92 iterations, with a root mean square value of 1.59 (presented in Fig. 6D; fit curves in Fig. S3), recovered the high resistivity associated with the Jalisco Block complex and showed that at ~8 km depth, the electrical resistivity decreases again (Fig. 6A). This transition is explained as an increase of the conductivity at greater depths or due to a loss of sensitivity of the model.

Additional inversions to test the sensitivity of important features were also performed. We calculated 3-D inversions considering a homogeneous medium of 250 Ohm-m at 0 masl, 2 mbsl, and 5 mbsl. The results obtained are presented in Figure S4.

4. SPATIAL DISTRIBUTION OF ANOMALOUS CONDUCTORS

The results are now analyzed as four plane views (Figs. 7A–7D). The figure displays the distribution of resistivity at different elevations. The arrows indicate the direction from the shallower (700 masl) to the deepest (350 mbsl). The 700 masl section (Fig. 7A) corresponds to a subsurface depth of ~600 m within the domain of the SPD. At this elevation, the anomalous low resistivity (<10 Ohm-m), correlates to the top of the altered cap layer as confirmed through the drilled wells in the SPD geothermal field. A similar anomaly, either in size and conductivity values, is observed south of the area in the surroundings of the Amado Nervo and Borbollón faults. However, neither of these resistivity anomalies exhibits a magnetic response (Fig. 7E). This absence of magnetic response, especially in the context of the SPD, directly confirms that the high electrical conductivity of the mineralized cap layer is not influenced by ferromagnesian or magnetite-rich minerals, which may be also the case for the southern Amado Nervo fault and Borbollón fault high conductivity anomaly.

At 350 masl, the high conductor associated with the SPD cap layer vanishes, and a new low resistivity zone appears at the center of the Compostela graben, more related with the Amado Nervo fault and Borbollón fault anomalous conductive zone rather than with to the SPD. Except for the SE (Fig. 7B), most of the area becomes highly resistive, unveiling compact unaltered granitic basement rocks. The outlined high magnetization zone only marginally matches the conductor. At mean sea level (Fig. 7C), the high conductivity anomaly has increased its size and

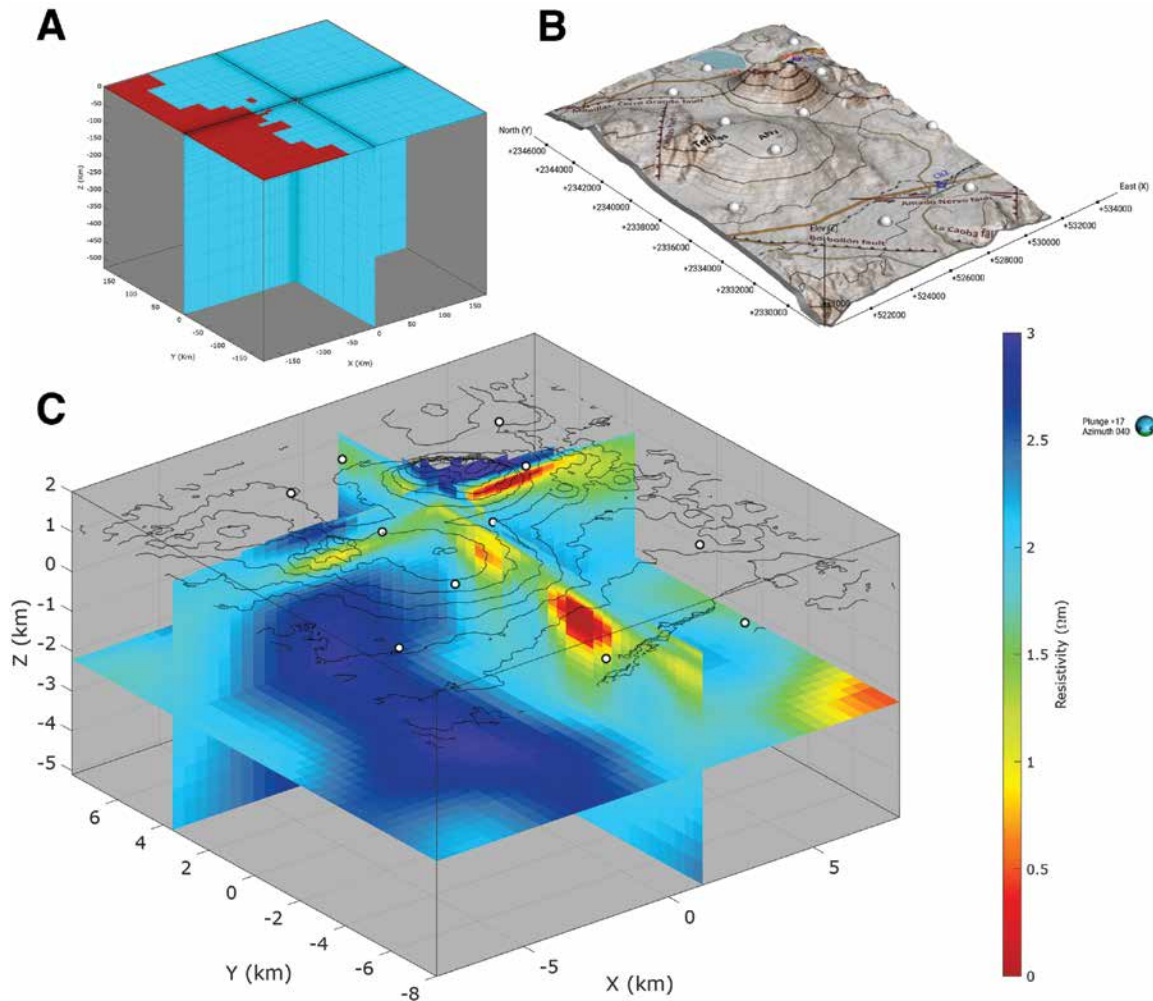


Figure 5. Starting model mesh (A) and 3-D inversion model (C) were obtained after 143 iterations and an average root mean square of 1.62; (B) is a perspective of topography. In (A) the red color corresponds to 0.3 Ohm-m of the ocean and the green color corresponds to the homogeneous half-space of 100 Ohm-m. The animation clip for 3-D inspection of the results can be found in the supplemental file (Movie.wmv; see text footnote 1).

Figura 5. Mallado inicial del modelo (A) y del modelo de la inversión 3D (C) obtenido tras 143 iteraciones, con un error cuadrático medio (RMS) promedio de 1,62; (B) muestra una perspectiva de la topografía. Además, en (A), el color rojo corresponde a la resistividad del océano (0,3 Ω -m) y el verde a la resistividad homogénea inicial (100 Ω -m). Como archivo suplementario se presenta un vídeo de animación 3D de los resultados obtenidos (Movie.wmv; véase la nota al pie del texto 1).

migrated southward, yet only a partial correlation with the high magnetic anomalies is observed. However, at 350 mbsl (Fig. 7D), most of the low resistivity anomalous zone can be correlated with a corresponding high magnetization anomalous zone within the Compostela graben extending toward the west of the studied area. The overall trend of the high conductivity zone and its magnetic response suggest that the conductor's nature at greater depths is either ferromagnesian and/or with high content of magnetic minerals. It also suggests upward migrations of conductive plumes from the center of the Compostela graben toward the faulted surface areas of the SPD (Milpillás-Cerro Grande fault), and the Amado Nervo-Borollón faults zone.

5. VERTICAL RESISTIVITY SECTIONS: DEPTHS AND FAULT STRUCTURES

Four vertical resistivity cross sections were extracted from the 3-D model for analysis of the structures and lithological distribution in the area, identified from the samples of the drilled wells in the SPD geothermal field (Fig. 2). Two profiles (P-2 and P-3) are parallel to each other and oriented N-S, and the other two (P-1 and P-4) are perpendicular to each other and oriented SW-NE and NW-SE, respectively. A common feature across all the resistivity depth sections is the occurrence of a shallow main low resistivity layer (<10 Ohm-m) that seems to be at 0 masl. The

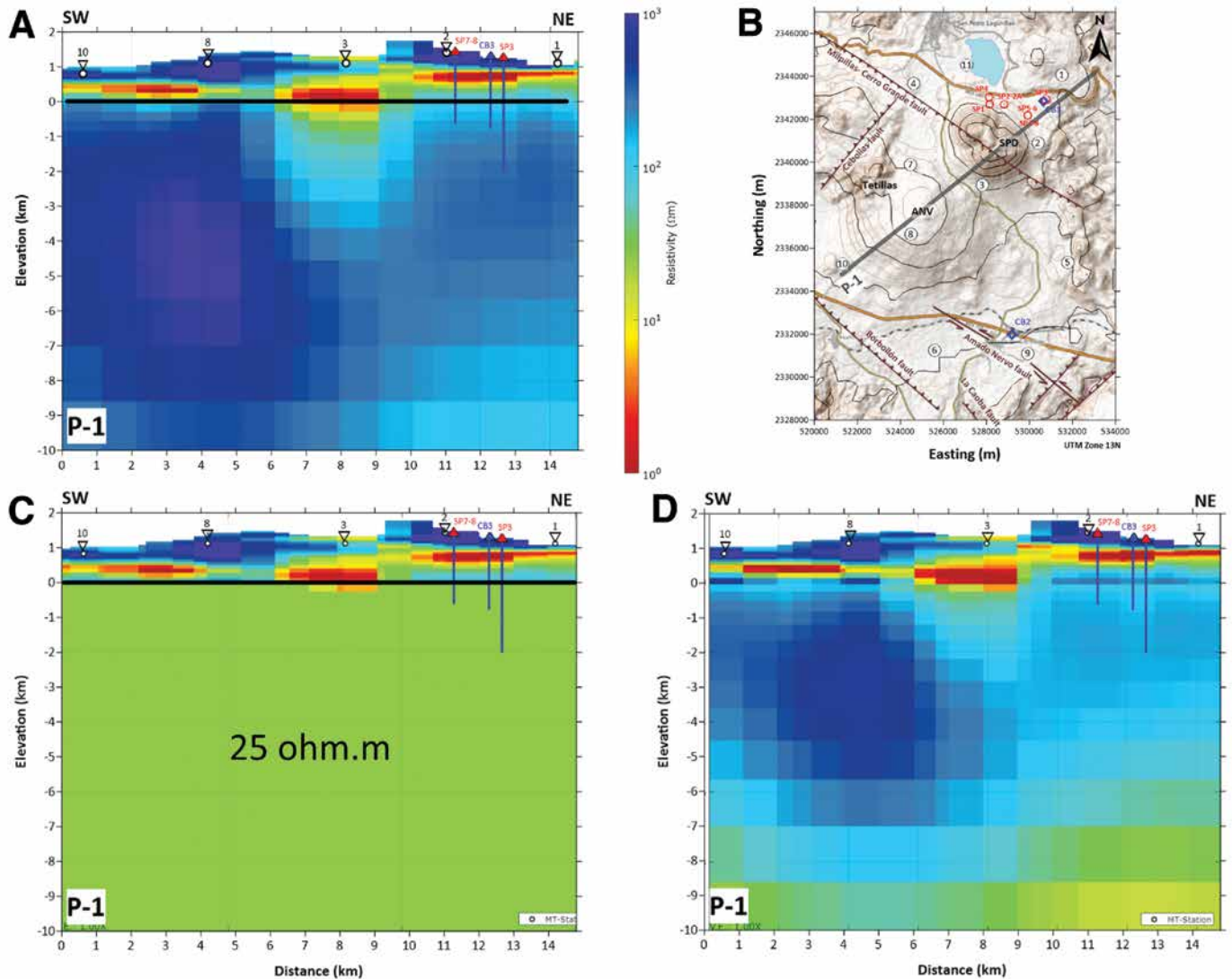


Figure 6. Resistivity model testing. (A) P-1 profile. (B) Modified version of Figure 2; result of preferred inversion resistivity model. (C) Profile P-1 with homogeneous electrical resistivity below 0 masl. (D) Result of constrained 3-D inversion according to (C). SPD—San Pedro Dome; ANV—Amado Nervo volcano.

Figura 6. Pruebas de sensibilidad del modelo. (A) Perfil P-1. (B) Versión modificada de Figura 2; de resistividades eléctricas resultante de la inversión 3D. (C) Perfil P-1 con resistividad eléctrica homogénea por debajo de 0 msnm. (D) resultado de la inversión 3D restringida según (C). SPD—Domo de San Pedro; ANV—Volcán Amado Nervo.

sections also reveal a change in the bulk resistivity of the granite basement from west to east, with higher conductivity and possible mineral alteration observed toward the east and south of the study area. According to the well's stratigraphy, the resistivities above ~150 Ohm-m, outlined with a solid white line in the sections of Figure 8, represent the contact surface of the granite basement rocks of the Jalisco Block complex, which provides a view of the depth and altered state of the granitic basal rocks across the area. The upward-pointing red arrows in the resistivity sections represent the most probable pathways for conductive fluids toward the surface, based on observed high resistivity gradients

and sudden changes in the cap layer conductor thickness. Dashed lines represent impermeable faults acting as hydrothermal barriers, while solid lines in the sections represent surface faults projected downward from available cartography of the zone (SGM, 2006a). The high resistivity areas (>1000 Ohm-m) to the west, along profiles 1 and 2, and the NW of profile 4, are interpreted as compact impermeable and denser granitic basement rocks consistent with the findings from drilled wells (Fig. 4). However, resistivities lower than ~150 Ohm-m are associated with more altered, partially permeable, and less dense granites, predominantly located to the east and southeast of the covered

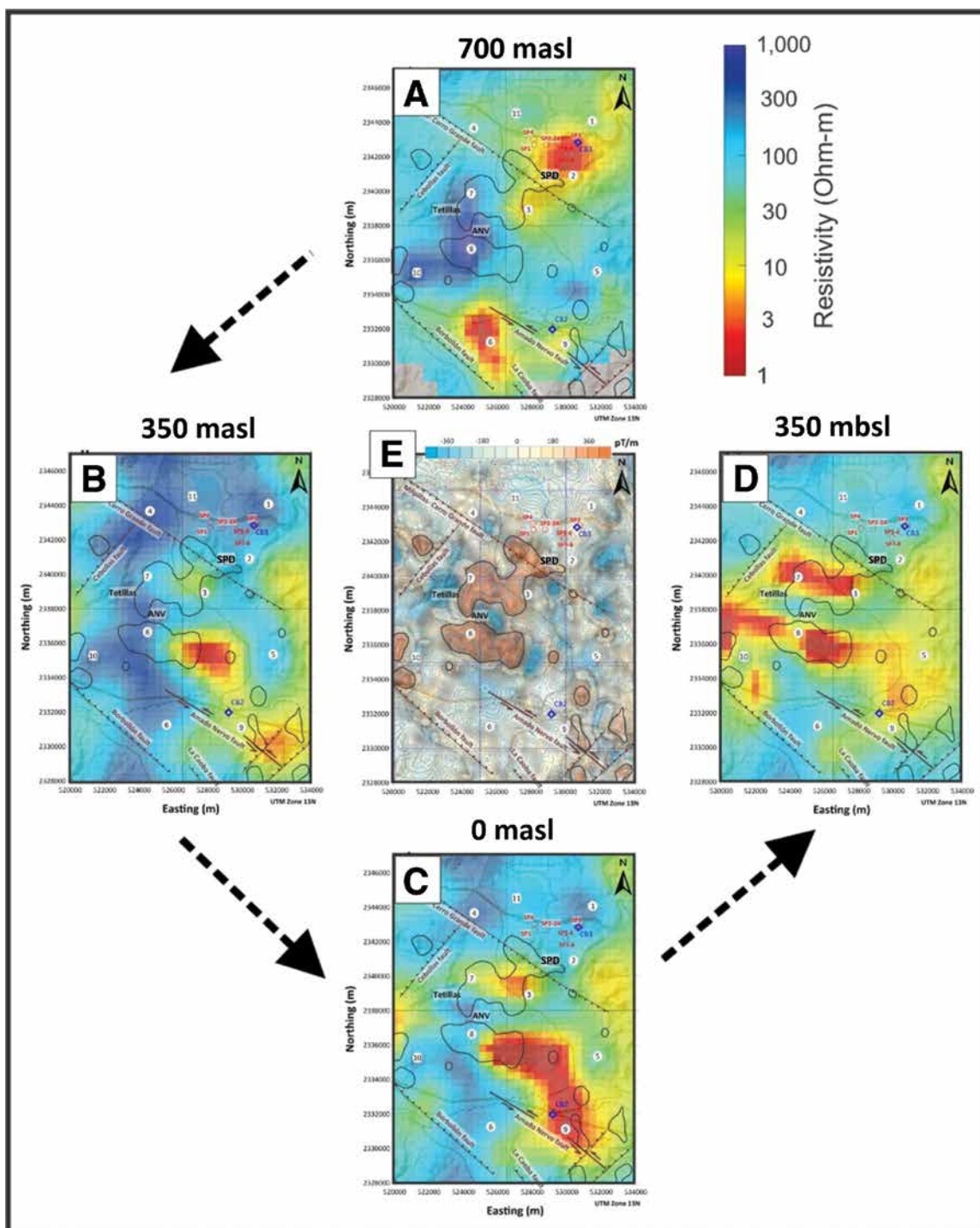


Figure 7. Resistivity maps obtained from the 3-D model at (A) 700 masl, (B) 350 masl, (C) 0 masl, and (D) 350 mbsl. Additionally, the map of the vertical derivative of the residual magnetic field reduced to the pole is included at the center of the figure for comparison (Servicio Geológico Mexicano, 2006b). The anomalies in this map are transformed lat-long dependent anomalies into magnetic anomalies centered in the producing bodies. The outline of the high magnetic anomaly zone is drawn on the resistivity charts for comparison. SPD—San Pedro Dome; ANV—Amado Nervo volcano.

Figura 7. Mapas de resistividad obtenidos del modelo 3D a: (A) 700 msnm., (B) 350 msnm., (C) 0 msnm y (D) -350 msnm. Adicionalmente, se incluye en el centro de la figura, para comparación, el mapa de la derivada vertical del campo magnético residual reducido al polo (Servicio Geológico Mexicano, 2006b). Las anomalías en este mapa son dependientes de la latitud y longitud centradas en los cuerpos productores. Para su comparación, se muestran los contornos de altos valores de la anomalía magnética en los mapas de resistividad. SPD—Domo San Pedro; ANV—Volcán Amado Nervo.

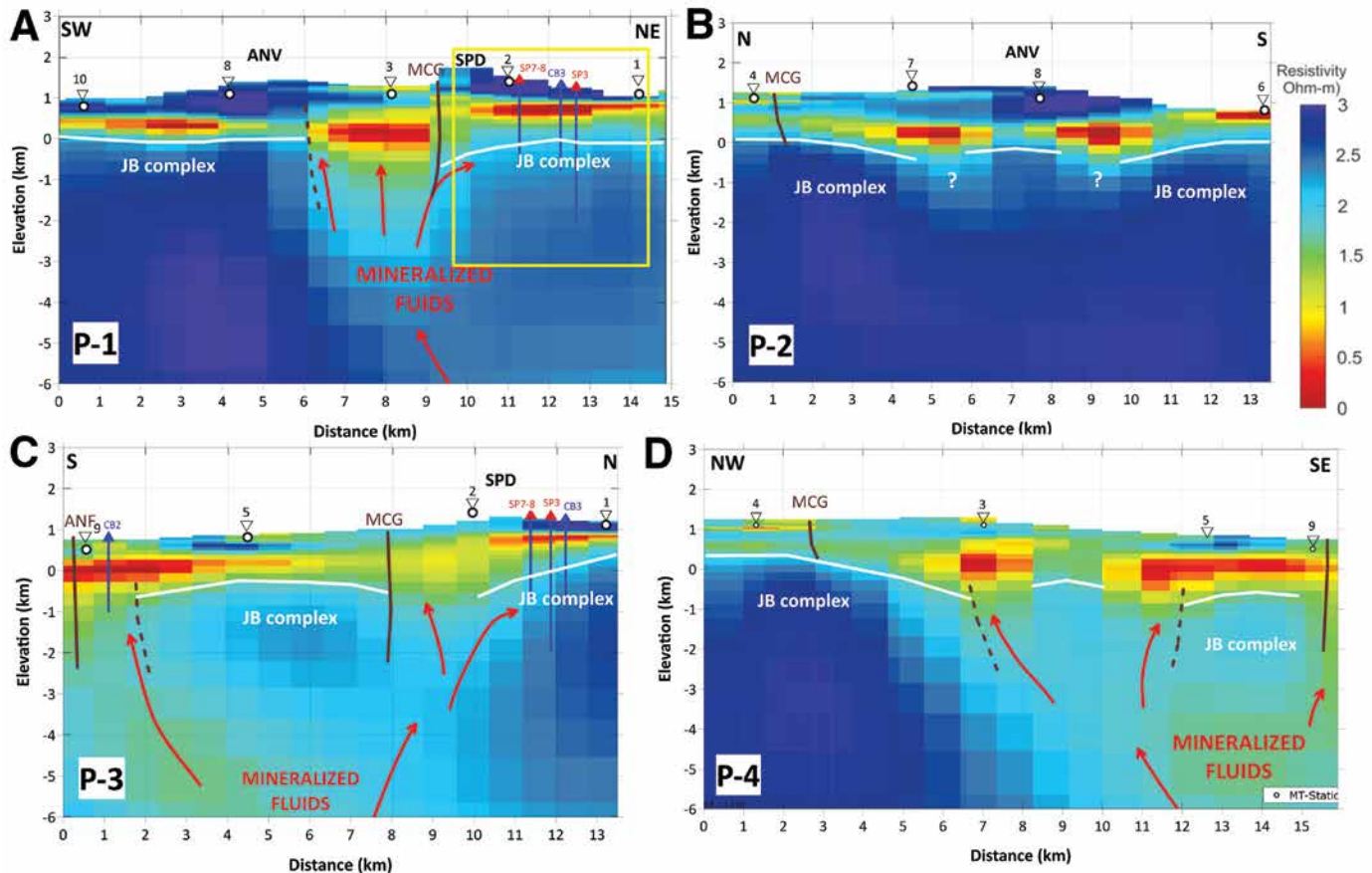


Figure 8. Vertical resistivity-depth profiles (Fig. 2) extracted from the cubed model (Movie.wmv; see text footnote 1) analyzed in terms of surface and inferred structures and calibrated lithology from samples of drilled wells within the San Pedro Dome (SPD) geothermal field. The solid white line shows the contact surface of the granite basement (Jalisco Block [JB] complex) extrapolated from the wells data using resistivity values >150 Ohm-m from the 1-D inversions. Upward-pointing red arrows represent most probable hot fluid paths toward the surface. Dash dotted brown lines are inferred faults, and solid brown lines represent the faults from the Servicio Geológico Mexicano map (SGM, 2006a). The yellow rectangle is the area of the P-1 that was used in Figure 4. ANV—Amado Nervo volcano; ANF—Amado Nervo fault; MCG—Milpillás—Cerro Grande fault.

Figura 8. Perfiles de resistividad vs Elevación (ver Fig. 2) extraídos del modelo cúbico 3D de la inversión (Movie.wmv; véase la nota al pie del texto 1), analizados en términos de las estructuras superficiales e inferidas y de la litología calibrada a partir de muestras de pozos perforados dentro del campo geotérmico Domo de San Pedro (SPD). La línea blanca continua muestra la superficie de contacto del basamento granítico (complejo JB) extrapolada a partir de los datos de los pozos utilizando valores de resistividad >150 Ω -m de las inversiones 1D. Las flechas rojas ascendentes representan las trayectorias más probables de fluidos calientes hacia la superficie. Las líneas marrones punteadas representan las fallas inferidas, y las continuas representan las fallas del Servicio Geológico Mexicano map (SGM, 2006a). El rectángulo amarillo corresponde al área del perfil P-1 que se usó en la Figura 4. ANV—Volcán Amado Nervo; AN—Falla de Amado Nervo; MCG—Falla de Milpillás—Cerro Grande.

area. We suggest that the lower bulk resistivity is the result of the induced fracturing by over-pressured fluids migrating from middle, or even lower crustal depths (e.g., Arzate et al., 2023) and percolating pervasively through a more permeable rock matrix.

6. DISCUSSION OF RESULTS

The correlation of the cross sections with the lithology reported from drilled wells allows the calibration of the low resistivity layer (<10 Ohm-m) with an ~ 500 -m-thick layer of argillized Miocene andesitic-dacitic rocks, considered the sealing cap of the reservoir. Argillization occurring within a

temperature range of 50 – 200 $^{\circ}\text{C}$ (Browne, 1978) produces electrically conductive clay layers of smectite and smectite-illite. Beneath this highly conductive clay layer, the resistivity increases to ~ 100 Ohm-m even though the temperature continues to rise. Temperatures exceeding 220 – 240 $^{\circ}\text{C}$ lead to propylitic alteration, mainly in the form of chlorite and epidote (e.g., Ussher et al., 2000), with a reported resistivity ranging from 10 to 100 Ohm-m (Wright et al., 1985; Pellerin et al., 1996). In the SPD geothermal system, the propylitic alteration is associated with an anomalous resistivity range of 30 – 80 Ohm-m, extending to greater depths in some sectors, and toward the eastern sector of the Compostela graben (Figs. 8 and 9). The propylitic

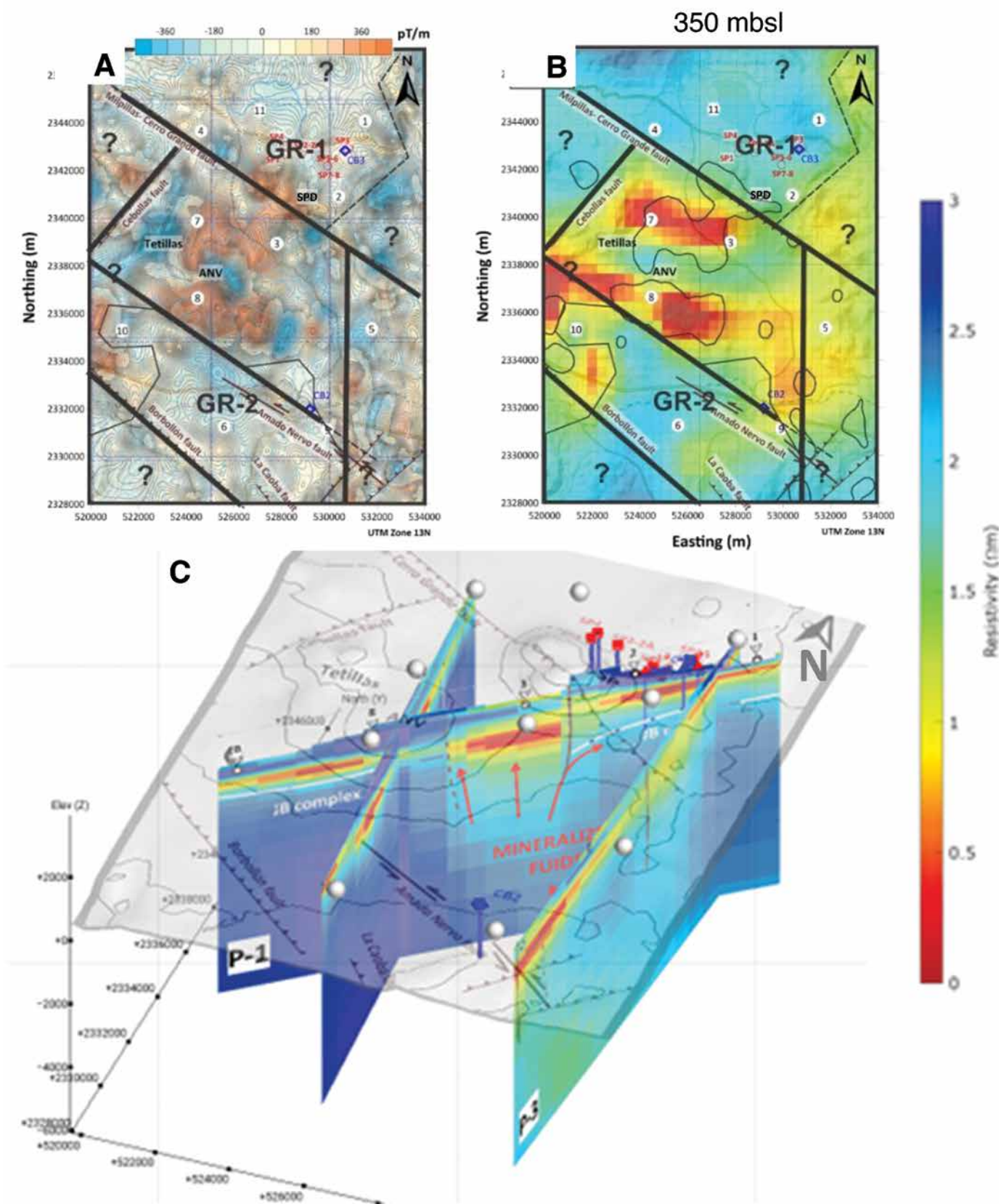


Figure 9.

alteration includes also hematization at higher depths and temperatures ranging from ~300 to 450 °C caused by oxidizing fluids interacting with the argillized andesitic rocks. Within the central area of the Compostela graben, the occurrence of hematization is indicated by the gradient of the residual of the total magnetic field ($\Delta[\text{RES}]/\text{TMF}$), shown at the center of Figure 7. The vertical gradient accentuates the presence of magnetite-rich mineralization, effectively highlighting magnetic anomalies underlying non-magnetic formations. This map broadly displays the concentration areas of magnetite ($\text{Fe}_2\text{Fe}_3+2\text{O}_4$) minerals and ferromagnesian silicate minerals ($\text{MgSO}_4\text{-FeSiO}_4$). This anomalous magnetic concentration widely coincides with the anomalous low resistivity zones (<10 Ohm-m) of Figure 7D, which corresponds to an approximate depth of 1800 m below the surface.

The lower part of the SPD hydrothermal system comprises the granitic basement (Fig. 8), characterized by resistivity values above ~150 Ohm-m in its most altered state, obtained from the Occam 1-D inversion of the determinant of Z of two MT soundings alongside two calibrating wells (SP7–8 and SP3). This value broadly corresponds to an equilibrium “flotation” line of 0 masl, suggesting the isostatic equilibrium of the Jalisco Block granitic

complex in this region. The lower bulk resistivity observed in the granite basement in the center and south of the Compostela graben implies the potential extension of the reservoir into these areas. This is assuming that the resistivity of the more porous mineralized fluid flow pathways, which are at the same time the storage place, is in the range of 30 to <100 Ohm-m, lower than the altered granite resistivity unit. The more porous zone extends to greater depths in certain sectors and toward the southeast of the Compostela graben, suggesting that the source of hot mineralized fluids of the regional geothermal system comes from the central and southeast region of the Compostela graben. In this context, the fault surface structures, such as the Milpillás–Cerro Grande fault (continuous dark line in Fig. 8), as well as those inferred from high resistivity gradients from the 3-D model (dashed dark lines), seem to act as impermeable barriers rather than as permeable pathways; fluids appear to follow a pre-faulted, more permeable pathway through the granite while avoiding the argillized, impermeable fault system of central Compostela graben inner faults.

In particular, given the results and previous arguments, we find a close relationship between the deep fault system of the SPD and the late Miocene faulting of the granitic basement rocks, which is particularly less dense toward the SE of the studied area (see Fig. 1). The reduction in density is naturally attributed to an increase in pore fluid within a fractured granite matrix, which consequently decreases the electrical resistivity. Figures 9A and 9B illustrate these arguments, where the thick lines on top of the mapped surface faults indicate that these outcropping structures are impermeable barriers made of argillized minerals. Additionally, we identify two distinct potential geothermal reservoirs (GR), separated by a confined zone of high conductivity and high magnetization. GR-1 is the SPD geothermal reservoir, bounded to the south by the Milpillás–Cerro Grande fault, and GR-2 represents a potential geothermal reservoir associated with the Amado Nervo volcano. A composite 3-D representation of the Compostela Graben geothermal system is presented in a video clip (Movie.wmv; see footnote 1) synthesizing the main results of this MT survey.

7. CONCLUSIONS

Wide-band magnetotelluric (MT) soundings have become the main tool in geothermal prospecting, not only because of the deeper exploration depths reached that allows heat source modeling, but also due to their outstanding resolution at high frequencies ($10^3\text{--}10^0$ Hz), offering a detailed near-surface more detailed prospecting.

The present study contributed to the understanding of the SPD active geothermal field and for the semi-regional geothermal system of the Compostela Graben. This geophysical work must be considered from the point of view of future monitoring of an active geothermal fields to observe and understand possible changes in physical proxies. The integrated model provides a useful and robust tomographic imagery of the SPD

Figure 9. Comparison between the vertical derivate of the residual magnetic field reduced to the pole map (A) provided by Servicio Geológico Mexicano (2006b) and the resistivity map obtained from the 3-D resistivity model at 350 mbsl (B). GR-1 represents the San Pedro Dome (SPD) potential geothermal reservoir, bounded to the south by the Milpillás–Cerro Grande (MCG) fault, and GR-2 represents a potential geothermal reservoir associated with the Amado Nervo volcano (ANV). (C) A composite model of the Compostela graben from the four interpreted resistivity profiles against the simplified topographic map of the studied area. This view provides a 3-D perspective of the upper 6–8 km of the regional geothermal system, highlighting the extrapolated surface faults (continuous lines) and inferred faults (dashed lines) as well as the proposed pathways of over-pressured mineralized fluids (red arrows). Both vertical and horizontal scales are the same. The animated 3-D clip of the resistivity cube is available as the Movie.wmv supplemental file (see text footnote 1). JB—Jalisco Block.

Figura 9. Comparación entre la derivada vertical del campo magnético residual reducido al polo (A) proporcionado por el Servicio Geológico Mexicano (2006b) y el mapa de planta de resistividad obtenido de la inversión 3D a -350 msnm (B). GR-1 representa el reservorio geotérmico potencial Domo de San Pedro (SPD), limitado al sur por la falla Milpillás–Cerro Grande (MCG), y GR-2 representa un reservorio geotérmico potencial asociado al Volcán Amado Nervo. (C) Modelo compuesto del graben de Compostela a partir de los cuatro perfiles de resistividad interpretados, sobre el mapa topográfico simplificado del área de estudio. Esta vista ofrece una perspectiva 3D de los 6–8 km superiores del sistema geotérmico regional, destacando las fallas superficiales extrapoladas (líneas continuas) y las fallas inferidas (líneas punteadas), así como las posibles rutas de los fluidos mineralizados sobre-presurizados (flechas rojas). Las escalas vertical y horizontal son las mismas. El vídeo 3D animado del cubo de resistividad está disponible como material suplementario, Movie.wmv (véase la nota al pie del texto 1). JB—Bloque de Jalisco.

hydrothermal system for further optimal and sustainable development of the region.

The calibrated structural subsurface model, derived from the 3-D inversion of MT data, and compared with the vertical gradient of the residual of the total magnetic field, allows the distinction between shallow and deeper magnetite-rich mineralized zones. These act as indicators of horizontal geothermal sealing cap of the reservoir or for the mineralized near-vertical fluids source paths. The resistivity images from the 3-D MT data inversion provide a proxy between the low resistivity zones and the upward-flowing hydrothermal fluids containing reactive minerals. At the same time, the results of this work enhance our understanding on the role of some of the main faults confining the SPD geothermal reservoir.

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